50 Essential Linux Commands for Efficient Command-Line Usage

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1. **ls** – Lists files and directories in the current path.

   👉 `ls -l`

2. **cd** – Changes the current working directory.

   👉 `cd /home/user`

3. **pwd** – Prints the current directory path.

   👉 `pwd`

4. **mkdir** – Creates a new directory.

   👉 `mkdir new_folder`

5. **rm** – Removes files or directories.

   👉 `rm file.txt` or `rm -r folder`

6. **cp** – Copies files or directories.

   👉 `cp file.txt /backup/`

7. **mv** – Moves or renames files/directories.

   👉 `mv oldname.txt newname.txt`

8. **cat** – Displays the content of a file.

   👉 `cat notes.txt`

9. **grep** – Searches for a pattern in files.

   👉 `grep "error" logfile.txt`

10. **chmod** – Changes file permissions.

    👉 `chmod 755 script.sh`

11. **chown** – Changes file owner.

    👉 `chown user file.txt`

12. **chgrp** – Changes group ownership.

    👉 `chgrp developers project/`

13. **sudo** – Runs a command as a superuser.

    👉 `sudo apt update`

14. **apt** – Manages packages on Debian-based systems.

    👉 `sudo apt install nmap`

15. **yum** – Manages packages on RPM-based systems.

    👉 `sudo yum install httpd`

16. **ps** – Displays running processes.

    👉 `ps aux`

17. **kill** – Terminates a process by PID.

    👉 `kill 1234`

18. **top** – Displays real-time system processes and usage.

    👉 `top`

19. **df** – Shows disk space usage.

    👉 `df -h`

20. **du** – Shows directory space usage.

    👉 `du -sh /var/log`

21. **man** – Displays the manual for a command.

    👉 `man ls`

22. **ssh** – Connects securely to a remote host.

    👉 `ssh [email protected]`

23. **scp** – Copies files between systems securely.

    👉 `scp file.txt [email protected]:/home/user/`

24. **tar** – Archives multiple files.

    👉 `tar -cvf backup.tar /home/user`

25. **gzip** – Compresses files using gzip.

    👉 `gzip file.txt`

26. **gunzip** – Decompresses gzip-compressed files.

    👉 `gunzip file.txt.gz`

27. **ping** – Tests network connectivity to a host.

    👉 `ping google.com`

28. **ifconfig** – Displays network interface configuration.

    👉 `ifconfig eth0`

29. **netstat** – Displays network connections and ports.

    👉 `netstat -tuln`

30. **wget** – Downloads files from the internet.

    👉 `wget https://example.com/file.zip`

31. **curl** – Transfers data from or to a server.

    👉 `curl https://api.github.com`

32. **history** – Shows previously used commands.

    👉 `history | grep ssh`

33. **find** – Searches files by name or attribute.

    👉 `find /home -name "report.txt"`

34. **locate** – Quickly finds files by name.

    👉 `locate passwd`

35. **tail** – Displays the last lines of a file.

    👉 `tail -n 10 log.txt`

36. **head** – Displays the first lines of a file.

    👉 `head -n 5 data.txt`

37. **sort** – Sorts lines in a file.

    👉 `sort names.txt`

38. **ssh-keygen** – Generates SSH key pairs.

    👉 `ssh-keygen -t rsa`

39. **sed** – Edits text in a stream or file.

    👉 `sed 's/error/success/g' log.txt`

40. **awk** – Processes and extracts data from text files.

    👉 `awk '{print $1, $3}' data.txt`

41. **cut** – Extracts specific columns or fields.

    👉 `cut -d':' -f1 /etc/passwd`

42. **wc** – Counts lines, words, and characters.

    👉 `wc -l file.txt`

43. **diff** – Compares two files line by line.

    👉 `diff file1.txt file2.txt`

44. **ln** – Creates hard or symbolic links.

    👉 `ln -s /usr/bin/python3 python`

45. **alias** – Creates shortcuts for commands.

    👉 `alias ll='ls -la'`

46. **echo** – Displays text or variables.

    👉 `echo $HOME`

47. **date** – Displays or sets the system date/time.

    👉 `date +"%Y-%m-%d %H:%M:%S"`

48. **clear** – Clears the terminal screen.

    👉 `clear`

49. **uptime** – Shows how long the system has been running.

    👉 `uptime`

50. **whoami** – Displays the current logged-in username.

👉 `whoami`
